

## SAFETY DATA SHEET

Revision Date 28-Dec-2021

Revision Number 4

### 1. Identification

**Product Name** Protein Gel Loading Dye,2X

**Cat No. :** AC611960000; AC611960010; AC611960050

**Synonyms** Mixture

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number** For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Carcinogenicity

Category 1B

#### Label Elements

**Signal Word**  
Danger

**Hazard Statements**  
May cause cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use

Do not handle until all safety precautions have been read and understood

Use personal protective equipment as required

**Response**

IF exposed or concerned: Get medical attention/advice

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

WARNING. Cancer - <https://www.p65warnings.ca.gov/>.

### 3. Composition/Information on Ingredients

| Component                        | CAS No    | Weight % |
|----------------------------------|-----------|----------|
| Water                            | 7732-18-5 | 80.0     |
| Glycerin                         | 56-81-5   | < 12.0   |
| Sodium lauryl sulfate            | 151-21-3  | < 5.0    |
| Proprietary Ingredient           | NA        | < 2.0    |
| Tris(2,3-dibromopropyl)phosphate | 126-72-7  | < 1.0    |

### 4. First-aid measures

|                                            |                                                                                          |
|--------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes.                       |
| <b>Inhalation</b>                          | Remove to fresh air.                                                                     |
| <b>Ingestion</b>                           | Do NOT induce vomiting.                                                                  |
| <b>Most important symptoms and effects</b> | No information available.                                                                |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                    |

### 5. Fire-fighting measures

|                                       |                          |
|---------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b> | No information available |
| <b>Flash Point</b>                    |                          |
| <b>Method -</b>                       | No information available |
| <b>Autoignition Temperature</b>       | No information available |
| <b>Explosion Limits</b>               |                          |
| <b>Upper</b>                          | No data available        |

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

None known.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
1

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

**6. Accidental release measures****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

No information available.

**7. Handling and storage****Handling**

Ensure adequate ventilation.

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place.

**8. Exposure controls / personal protection****Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Component | ACGIH TLV | OSHA PEL                                                                                                                           | NIOSH IDLH | Mexico OEL (TWA)          |
|-----------|-----------|------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------|
| Glycerin  |           | (Vacated) TWA: 10 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> |            | TWA: 10 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Liquid                   |
| Appearance                             | Purple                   |
| Odor                                   | Odorless                 |
| Odor Threshold                         | No information available |
| pH                                     |                          |
| Melting Point/Range                    | No data available        |
| Boiling Point/Range                    |                          |
| Flash Point                            |                          |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | Soluble in water         |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

|                                  |                                            |
|----------------------------------|--------------------------------------------|
| Reactive Hazard                  | None known, based on information available |
| Stability                        | Stable under normal conditions.            |
| Conditions to Avoid              | Incompatible products.                     |
| Incompatible Materials           | Strong oxidizing agents                    |
| Hazardous Decomposition Products | None under normal use conditions           |
| Hazardous Polymerization         | Hazardous polymerization does not occur.   |
| Hazardous Reactions              | None under normal processing.              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**Oral LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Vapor LC50** Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component                        | LD50 Oral                 | LD50 Dermal                 | LC50 Inhalation                           |
|----------------------------------|---------------------------|-----------------------------|-------------------------------------------|
| Water                            | -                         | -                           | -                                         |
| Glycerin                         | 12600 mg/kg ( Rat )       | > 10 g/kg ( Rabbit )        | > 2.75 mg/L/4h ( Rat )(mist)              |
| Sodium lauryl sulfate            | LD50 = 1288 mg/kg ( Rat ) | LD50 = 200 mg/kg ( Rabbit ) | LC50 > 3900 mg/m <sup>3</sup> ( Rat ) 1 h |
| Tris(2,3-dibromopropyl)phosphate | LD50 = 810 mg/kg ( Rat )  | LD50 = 17.6 g/kg ( Rabbit ) | Not listed                                |

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                        | CAS No    | IARC       | NTP                    | ACGIH      | OSHA       | Mexico     |
|----------------------------------|-----------|------------|------------------------|------------|------------|------------|
| Water                            | 7732-18-5 | Not listed | Not listed             | Not listed | Not listed | Not listed |
| Glycerin                         | 56-81-5   | Not listed | Not listed             | Not listed | Not listed | Not listed |
| Sodium lauryl sulfate            | 151-21-3  | Not listed | Not listed             | Not listed | Not listed | Not listed |
| Proprietary Ingredient           | NA        | Not listed | Not listed             | Not listed | Not listed | Not listed |
| Tris(2,3-dibromopropyl)phosphate | 126-72-7  | Group 2A   | Reasonably Anticipated | Not listed | X          | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects,both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component             | Freshwater Algae                                                                                                                                                                                                                        | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                                                                                 | Microtox                                                                                                                                                      | Water Flea                            |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Glycerin              | Not listed                                                                                                                                                                                                                              | LC50: 51 - 57 mL/L, 96h static (Oncorhynchus mykiss)                                                                                                                                                                                                                                                                                                                                                                                            | Not listed                                                                                                                                                    | Not listed                            |
| Sodium lauryl sulfate | EC50: 3.59 - 15.6 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: = 117 mg/L, 96h (Pseudokirchneriella subcapitata)<br>EC50: 30 - 100 mg/L, 96h (Desmodesmus subspicatus)<br>EC50: = 53 mg/L, 72h (Desmodesmus subspicatus) | 1.31 mg/L LC50 96 h<br>9.9-20.1 mg/L LC50 96 h<br>4.5 mg/L LC50 96 h<br>4.62 mg/L LC50 96 h<br>7.97 mg/L LC50 96 h<br>10.2-22.5 mg/L LC50 96 h<br>10.8-16.6 mg/L LC50 96 h<br>13.5-18.3 mg/L LC50 96 h<br>15-18.9 mg/L LC50 96 h<br>22.1-22.8 mg/L LC50 96 h<br>4.06-5.75 mg/L LC50 96 h<br>4.2-4.8 mg/L LC50 96 h<br>4.3-8.5 mg/L LC50 96 h<br>5.8-7.5 mg/L LC50 96 h<br>6.2-9.6 mg/L LC50 96 h<br>8-12.5 mg/L LC50 96 h<br>4.2 mg/L LC50 96 h | = 0.46 mg/L EC50 Photobacterium phosphoreum 30 min<br>= 0.72 mg/L EC50 Photobacterium phosphoreum 15 min<br>= 1.19 mg/L EC50 Photobacterium phosphoreum 5 min | EC50: = 1.8 mg/L, 48h (Daphnia magna) |

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

| Component             | log Pow |
|-----------------------|---------|
| Glycerin              | -1.76   |
| Sodium lauryl sulfate | 1.6     |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                                   | RCRA - U Series Wastes | RCRA - P Series Wastes |
|---------------------------------------------|------------------------|------------------------|
| Tris(2,3-dibromopropyl)phosphate - 126-72-7 | U235                   | -                      |

### 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

### 15. Regulatory information

#### United States of America Inventory

| Component                        | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|----------------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Water                            | 7732-18-5 | X    | ACTIVE                                        | -                           |
| Glycerin                         | 56-81-5   | X    | ACTIVE                                        | -                           |
| Sodium lauryl sulfate            | 151-21-3  | X    | ACTIVE                                        | -                           |
| Proprietary Ingredient           | NA        | -    | -                                             | -                           |
| Tris(2,3-dibromopropyl)phosphate | 126-72-7  | X    | ACTIVE                                        | S                           |

#### Legend:

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

S - Indicates a substance that is identified in a proposed or final Significant New Use Rule.

#### TSCA 12(b) - Notices of Export

| Component                        | CAS No   | TSCA 12(b) - Notices of Export |
|----------------------------------|----------|--------------------------------|
| Tris(2,3-dibromopropyl)phosphate | 126-72-7 | Section 5                      |

#### International Inventories

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                        | CAS No    | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|----------------------------------|-----------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Water                            | 7732-18-5 | X   | -   | 231-791-2 | X     | X    |      | X    | X     | KE-35400 |
| Glycerin                         | 56-81-5   | X   | -   | 200-289-5 | X     | X    | X    | X    | X     | KE-29297 |
| Sodium lauryl sulfate            | 151-21-3  | X   | -   | 205-788-1 | X     | X    | X    | X    | X     | KE-21884 |
| Proprietary Ingredient           | NA        | -   | -   | -         | -     | -    |      | -    | -     | -        |
| Tris(2,3-dibromopropyl)phosphate | 126-72-7  | -   | -   | 204-799-9 | -     | X    | X    | X    | X     | KE-34800 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### U.S. Federal Regulations

**SARA 313** Not applicable

| Component                        | CAS No   | Weight % | SARA 313 - Threshold Values % |
|----------------------------------|----------|----------|-------------------------------|
| Tris(2,3-dibromopropyl)phosphate | 126-72-7 | < 1.0    | 0.1                           |

**SARA 311/312 Hazard Categories** See section 2 for more information

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

**CERCLA** Not applicable

| Component                        | Hazardous Substances RQs | CERCLA EHS RQs |
|----------------------------------|--------------------------|----------------|
| Tris(2,3-dibromopropyl)phosphate | 10 lb                    | -              |

**California Proposition 65** This product contains the following Proposition 65 chemicals.

| Component                        | CAS No   | California Prop. 65 | Prop 65 NSRL | Category   |
|----------------------------------|----------|---------------------|--------------|------------|
| Tris(2,3-dibromopropyl)phosphate | 126-72-7 | Carcinogen          | 0.3 µg/day   | Carcinogen |

**U.S. State Right-to-Know Regulations** Not applicable

| Component                        | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|----------------------------------|---------------|------------|--------------|----------|--------------|
| Water                            | -             | -          | X            | -        | -            |
| Glycerin                         | X             | X          | X            | -        | X            |
| Tris(2,3-dibromopropyl)phosphate | X             | X          | X            | X        | -            |

#### U.S. Department of Transportation

Reportable Quantity (RQ): Y  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

#### Other International Regulations

**Mexico - Grade** No information available

#### Authorisation/Restrictions according to EU REACH

| Component                        | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Tris(2,3-dibromopropyl)phosphate | -                                                                   | Use restricted. See item 4. (see link for restriction details)                | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

#### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Water     | 7732-18-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

|                                  |          |                |                |                |                |
|----------------------------------|----------|----------------|----------------|----------------|----------------|
| Glycerin                         | 56-81-5  | Listed         | Not applicable | Not applicable | Not applicable |
| Sodium lauryl sulfate            | 151-21-3 | Listed         | Not applicable | Not applicable | Not applicable |
| Proprietary Ingredient           | NA       | Not applicable | Not applicable | Not applicable | Not applicable |
| Tris(2,3-dibromopropyl)phosphate | 126-72-7 | Not applicable | Not applicable | Not applicable | Not applicable |

| Component                        | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water                            | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Glycerin                         | 56-81-5   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Sodium lauryl sulfate            | 151-21-3  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Proprietary Ingredient           | NA        | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Tris(2,3-dibromopropyl)phosphate | 126-72-7  | Not applicable                                                                            | Not applicable                                                                           | X                          | Not applicable                     |

## 16. Other information

**Prepared By** Regulatory Affairs  
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Email: EMSDS.RA@thermofisher.com

**Revision Date** 28-Dec-2021

**Print Date** 28-Dec-2021

**Revision Summary** This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**